

DAY 4: GPUS AND COOLING SYSTEMS

Practical Assignment

Topic: GPU and Cooling System

1. Introduction to GPU

GPU stands for **Graphics Processing Unit**.

A GPU is a computer component mainly used to process images, videos, games, and other graphics.

GPU is used for:

- Playing games
- Watching and streaming high-quality videos
- Video editing
- 3D designing
- Graphic designing
- AI and other graphics applications

Two types of GPU:

1. Integrated GPU

- Built into the processor.
- Usually shares system memory.
- Good for normal computer work and basic graphics.

2. Dedicated GPU

- A separate graphics card installed in the computer.
- Has its own graphics memory.
- Better for gaming, video editing, and heavy graphics.

TASK 1: IDENTIFY THE GPU

I checked the GPU installed in the computer.

GPU Information	Details
Brand	NVIDIA
Model	GeForce RTX 3060
Type	Dedicated GPU
Graphics Memory	12 GB
Main Use	Gaming and graphics processing

About the GPU: The NVIDIA GeForce RTX 3060 is a dedicated graphics card. It has its own graphics memory and can handle heavy graphics applications.

GPU Photo: Paste a clear GPU photo or screenshot here.

2. What is a Cooling System?

When a computer is working, its components produce heat. A cooling system removes this heat and helps keep the GPU at a safe temperature.

Main purpose: Remove heat + Reduce temperature + Protect the GPU

TASK 2: IDENTIFY THE COOLING SYSTEM

The GPU uses an **air-cooling system**. Air cooling mainly uses a fan and heatsink to remove heat.

Main Cooling Components

- 1. Fan:** Moves air around the GPU and helps remove hot air.
- 2. Heatsink:** Absorbs heat from the GPU and spreads it over a larger area.
- 3. Thermal Paste:** Helps transfer heat from the GPU chip to the heatsink.
- 4. Airflow:** Moves hot air away and brings cooler air into the computer.

How Air Cooling Works:

GPU produces heat → Thermal paste transfers heat → Heatsink absorbs heat → Fan moves air → Heat is removed

Cooling System Photo: Paste a clear photo or screenshot here. Label Fan, Heatsink, GPU, and Airflow.

3. Types of GPU Cooling

A. Air Cooling

Air cooling uses a fan, heatsink, thermal paste, and airflow.

Simple working: Heat → Heatsink → Fan → Hot air removed

Advantages: Less expensive, easy to install, easy to maintain, no liquid leakage, and good for normal gaming PCs.

B. Liquid Cooling

Liquid cooling uses special coolant to transfer heat away from the component. It normally includes a pump, coolant, tubes, water block, radiator, and fans.

Simple working: Heat → Coolant → Radiator → Fan → Heat removed

Advantages: Very effective cooling and useful for high-performance systems.

Disadvantages: More expensive, more complicated, more components, and possible leakage risk.

TASK 3: AIR COOLING VS LIQUID COOLING

Feature	Air Cooling	Liquid Cooling
Cooling Method	Uses air	Uses liquid/coolant
Main Parts	Fan and heatsink	Pump, coolant, tubes and radiator

Feature	Air Cooling	Liquid Cooling
Cost	Usually cheaper	Usually more expensive
Cooling	Good cooling	Very effective cooling
Installation	Easier	More complicated
Maintenance	Easy	More maintenance may be needed
Noise	Fans can make noise	Pump and fans can make noise
Leakage Risk	No liquid leakage	Small possibility of leakage
Best For	Normal gaming and everyday use	High-performance systems
Reliability	Simple and reliable	More complex because it has more parts

4. Difference in Simple Words

Air Cooling: Fan + Heatsink = Cooling

Simple, cheaper, easy to maintain, and easy to install.

Liquid Cooling: Coolant + Pump + Radiator = Cooling

More powerful, more expensive, more complicated, and suitable for high-performance systems.

TASK 4: GAMING PC FOR IPL STREAMING AND HEAVY GAMES

My Choice: AIR COOLING

If I build a gaming PC for streaming IPL matches and playing graphics-heavy games, I would choose air cooling for the GPU. Air cooling is simple, affordable, and easy to maintain. A good GPU fan and heatsink can provide enough cooling for gaming and streaming. It also does not have the extra pump, tubes, and coolant used in liquid cooling.

My Reasons:

- 1. Affordable:** Air cooling usually costs less.
- 2. Easy to Install:** It is easier to install than liquid cooling.
- 3. Easy to Maintain:** Cleaning the fan and heatsink is generally simple.
- 4. No Liquid Leakage:** There is no coolant or tubing, so there is no liquid leakage concern.

IMPORTANT POINTS TO REMEMBER

GPU = Graphics Processing Unit

It processes Games + Images + Videos + Graphics

Air Cooling: Fan + Heatsink = Air Cooling

Liquid Cooling: Coolant + Pump + Radiator = Liquid Cooling

FINAL CONCLUSION

A GPU is an important part of a computer because it processes graphics and helps run games and other graphics-heavy applications. Since a GPU produces heat while working, it needs a proper cooling

system. Air cooling uses a fan and heatsink, while liquid cooling uses coolant, a pump, and a radiator. For a normal gaming and streaming PC, I would choose air cooling because it is simple, affordable, reliable, and easy to maintain.